

Skills Summary

Volumes from Known Cross Sections and Cylindrical Shells

Use this reference while completing your worksheet or when studying.

Skill 1 — Slicing into known cross sections

One idea: $V = \int_a^b A(x) dx$, where $A(x)$ is the area of the slice at position x . The base region only tells you the *side* $s(x)$ of that slice; the shape of the cross section turns it into an area.

- square on the base: $A = s^2$ semicircle on the base (diameter s): $A = \frac{\pi}{8}s^2$
- equilateral triangle on the base: $A = \frac{\sqrt{3}}{4}s^2$

If lengths are in centimetres, A is in square centimetres and V is in cubic centimetres.

Example: Base: region under $y = 2x$ from $x = 0$ to $x = 3$; cross sections perpendicular to the x -axis are squares. Find V .

Step 1. The slices are named by x , so integrate in x : read the side of the square off the base, then square it to get the area.

Step 2. The side runs from the axis to the line, so $s = 2x$ and $A(x) = 4x^2$; then $V = \int_0^3 4x^2 dx = \left[\frac{4}{3}x^3\right]_0^3$.
 $\Rightarrow V = 36$

Try it: Base: region under $y = 3x$ from $x = 0$ to $x = 2$; square cross sections. Find V .

check: $V = 24$

Skill 2 — Cylindrical shells

Shell formula: $V = \int_a^b 2\pi (\text{radius})(\text{height}) dx$. For a region revolved about the y -axis the radius is x and the height is the function value, so the integral stays in x and the curve needs no rewriting. Think of unrolling each thin cylinder into a rectangle of width $2\pi r$, height h , thickness dx .

Example: The region under $y = x$ from $x = 0$ to $x = 2$ is revolved about the y -axis. Find V .

Step 1. Revolving about a vertical axis with the region described by y in terms of x makes shells the setup that needs no algebra.

Step 2. Radius x , height x , so $V = \int_0^2 2\pi x \cdot x dx = 2\pi \left[\frac{x^3}{3}\right]_0^2$.
 $\Rightarrow V = \frac{16\pi}{3} \approx 16.755$

Try it: The region under $y = x$ from $x = 0$ to $x = 3$, revolved about the y -axis.

check: $V \approx 56.549 = 18\pi$

Skill 3 — Radius when the axis moves

The radius is a distance, not a coordinate. For revolution about the vertical line $x = k$ with the region to the *left* of it, the shell at position x is $k - x$ away; with the region to the *right*, it is $x - k$. Check an endpoint before integrating: the radius must be largest for the shell furthest from the axis and zero for a shell on it.

Example: The region under $y = x^2$ from $x = 0$ to $x = 2$ is revolved about $x = 3$. Find V .
Step 1. The axis lies to the right of the region, so the shell radius is the gap $3 - x$; the height is still the function value.

Step 2. $V = \int_0^2 2\pi(3 - x)x^2 dx = 2\pi \left[x^3 - \frac{x^4}{4} \right]_0^2 = 2\pi(8 - 4).$

$\Rightarrow V = 8\pi \approx 25.133$

Try it: The same region under $y = x^2$ on $[0, 2]$, now revolved about $x = 5$.

check: $V \approx \frac{8}{5}\pi = 5.026548245743669$